

# Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: INJONGO | Context: Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS

**Output Ontology definition:** Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

MODERATE-priority dimension with concrete structural gaps. The 23-slot ontology was rigorously developed with merge consolidation [\[Q54, Q55, Q119\]](#) and includes deployment-relevant types (BANK\_NAME, PAYMENT\_COMPANY, BILL\_TYPE, MONEY, ACCOUNT\_TYPE, CITY\_OR\_PROVINCE, PLACE\_NAME, DATE, TIME). However, the dimension fails on critical deployment requirements: dataset inspection confirms (D19) that no METER\_ID, UTILITY\_ACCOUNT\_NUMBER, STS\_TOKEN, KEBELE, WOREDA, or sub-city slot types exist in any of 19 Amharic examples, and these are core to EEU prepaid top-up and AAWSA billing workflows. Nine 'infrequent slots' were eliminated [\[Q52\]](#) without documented utility-deployment justification. Cross-lingual annotation choices show ontological ambiguity: SNEL (DRC electricity utility) is annotated as PAYMENT\_COMPANY (D28), conflating utility provider with payment channel — relevant if EEU/AAWSA appear similarly. The deployment elicitation flags this as MODERATE priority because BANK\_NAME/PAYMENT\_COMPANY/BILL\_TYPE provide partial coverage, but the missing types are non-negotiable for the core workflow.

| ID   | Evidence                                                                                                                                                                                                                                            |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q52  | "Consequently, nine infrequent slots from NATIONALITY through PLUG TYPE were eliminated." (p.4)                                                                                                                                                     |
| Q54  | "Appendix A.2 contains all 34 slot types selected." (p.4)                                                                                                                                                                                           |
| Q55  | "This merging process resulted in a reduction from 34 to 23 slot types." (p.5)                                                                                                                                                                      |
| Q119 | "After merging similar or low-frequency types during data preprocessing in Section 3.3, it was reduced to 23 distinct slot types as shown in the "Final Merged Type" column." (p.13)                                                                |
| D19  | በወር ለውጥ ስንት ነው መክፈል የሚገባኝ? — No METER_ID, UTILITY_ACCOUNT_NUMBER, or STS_TOKEN slots present anywhere in Amharic sample                                                                                                                             |
| D28  | "Niongo ya snel ezali 2000 FC Yebisa ngai soki nakosi" — PAYMENT_COMPANY: snel, MONEY: 2000 FC — Lingala: "SNEL bill is 2000 FC, tell me if I've paid" — SNEL is the DRC national electricity utility; utility company annotated as PAYMENT_COMPANY |

## Strengths

- 23 slot types include deployment-relevant categories: BANK\_NAME, PAYMENT\_COMPANY, BILL\_TYPE, MONEY, ACCOUNT\_TYPE, CITY\_OR\_PROVINCE, PLACE\_NAME, DATE, TIME, NUMBER [\[Q144\]](#)
- Telebirr fits PAYMENT\_COMPANY (D4) and BILL\_TYPE accommodates electricity (D1) — partial coverage of deployment slot space
- Annotation guide provides explicit disambiguation rules (e.g., temporal: <1 day → TIME; ≥1 day → DATE [\[Q153, Q154\]](#); bank name excludes 'bank' unless attached [\[Q152\]](#))
- Slot consolidation 34→23 was driven by frequency analysis [\[Q51, Q119\]](#), reducing label noise

| ID   | Evidence                                                                                                                                                                             |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q51  | "We decided to exclude slot entities appearing less than 500 times across all languages (after MUSIC_GENRE in the figure)." (p.4)                                                    |
| Q119 | "After merging similar or low-frequency types during data preprocessing in Section 3.3, it was reduced to 23 distinct slot types as shown in the "Final Merged Type" column." (p.13) |

|             |                                                                                                                                                                                                                                                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q144</b> | "# Named Entities Types to Identify ACCOUNT_TYPE, ARTIST_NAME, BANK_NAME, BILL_TYPE, CALENDAR_EVENT, CITY_OR_PROVINCE, COUNTRY, CURRENCY, DATE, DISH_OR_FOOD, HOTEL_NAME, LANGUAGE_NAME, MEAL_PERIOD, MONEY, MUSIC_GENRE, NUMBER, PAYMENT_COMPANY, PERSONAL_NAME, PLACE_NAME, RESTAURANT_NAME, SHOPPING_ITEM, SONG_NAME, TIME" (p.19) |
| <b>Q152</b> | "When annotating Bank names, you do not need to include "bank" unless it is attached to the bank name, like seen above, with Ecobank." (p.21)                                                                                                                                                                                         |
| <b>Q153</b> | "Anything that is less than one day should be annotated as TIME and not DATE, as seen in the above examples." (p.21)                                                                                                                                                                                                                  |
| <b>Q154</b> | "Anything that is more than one day must be annotated as DATE and not time, as seen above" (p.21)                                                                                                                                                                                                                                     |
| <b>D1</b>   | "በወር ለመብራት ስንት ነው መክፈል የሚገባኝ?" — BILL_TYPE: ለመብራት — "How much do I need to pay for electricity per month?" — electricity utility bill query in Amharic                                                                                                                                                                                |
| <b>D4</b>   | "በቴሌብር የብድር አካውንቱ ላይ አሁን ያለብኝ ቀሪ ሂሳብ ምን ያህል ነው።" — PAYMENT_COMPANY: ቴሌብር — "What is the remaining balance on my Telebirr loan account?" — Telebirr named as PAYMENT_COMPANY                                                                                                                                                           |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                          |
|-------------|--------------------------------------------------------------|
| <b>OO-1</b> | Review output label categories for regional relevance        |
| <b>OO-2</b> | Identify missing regionally specific categories              |
| <b>OO-3</b> | Flag categories encoding non-regional values                 |
| <b>OO-4</b> | Consider stakeholder-driven taxonomy redesign                |
| <b>OO-5</b> | Document taxonomy issues harming structural/content validity |

### Checklist responses:

**OO-1:** Deployment-relevant present categories: BANK\_NAME, PAYMENT\_COMPANY, BILL\_TYPE, MONEY, ACCOUNT\_TYPE, CITY\_OR\_PROVINCE, PLACE\_NAME, DATE, TIME, NUMBER [Q144].

**OO-2:** Confirmed missing for Ethiopian utility deployment (D19): METER\_ID, UTILITY\_ACCOUNT\_NUMBER, STS\_TOKEN (20-digit prepaid token), KEBELE, WOREDA, sub-city, NEIGHBORHOOD\_NAME (informal Addis Ababa toponyms). The annotation guide requires explicit named entities for PLACE\_NAME [Q156], so informal kebele/neighborhood references may not even fit existing types cleanly.

**OO-3:** Some category encodings reflect non-Ethiopian assumptions: PAYMENT\_COMPANY conflated with utility provider (D28: SNEL); SHOPPING\_ITEM, MUSIC\_GENRE, ARTIST\_NAME, DISH\_OR\_FOOD, MEAL\_PERIOD relevant only to non-utility domains. Nine slots (NATIONALITY through PLUG TYPE) were eliminated [Q52] — without justification of trade-off against utility-specific slots.

**OO-4:** Stakeholder-driven taxonomy redesign is required. EEU customer service agents and AAWSA billing staff should define slot schemas for METER\_ID format, account-number format, kebele granularity, and STS token recognition.

**OO-5:** Structural validity issue: deployment-critical decision boundaries (meter ID extraction, STS token recognition, kebele resolution) cannot be expressed in the 23-slot taxonomy, so benchmark performance does not reflect deployment task structure. Content validity issue: 9 missing slot types are deployment-critical.

| ID          | Evidence                                                                                                                                                                                                                                                                                                                              |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| <b>Q156</b> | "With this entity, only annotate if entity is named explicitly, e.g Name of airport, museum or mall is not and nt just "mall", "airport" etc" (p.24)                                                                                                                                                                                  |

|            |                                                                                                                                                                                                                                                     |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D19</b> | በወር ለመብራት ስንት ነው መክፈል የሚገባኝ? — No METER_ID, UTILITY_ACCOUNT_NUMBER, or STS_TOKEN slots present anywhere in Amharic sample                                                                                                                           |
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## Information Gaps

- EEU and AAWSA's actual slot schemas for meter IDs, account numbers, and addresses (not in public sources)
- Whether any pre-merge slot types (among the dropped 9) would have been useful for utility deployment

## Requires Expert Verification

- EEU billing system's canonical slot taxonomy and how informal Amharic place references should be normalized
- AAWSA's account-number and address conventions

## Remediation

**Gap 1: 23-slot ontology lacks METER\_ID, UTILITY\_ACCOUNT\_NUMBER, STS\_TOKEN, KEBELE, WOREDA, sub-city, and NEIGHBORHOOD\_NAME types — confirmed absent in dataset (D19).**

**Recommendation:** Co-design an extended slot schema with EEU and AAWSA billing staff; add at minimum METER\_ID, UTILITY\_ACCOUNT\_NUMBER, STS\_TOKEN, and KEBELE; re-annotate the existing Amharic split or annotate a deployment-specific augmentation under the extended schema.

| ID         | Evidence                                                                                                                  |
|------------|---------------------------------------------------------------------------------------------------------------------------|
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